Model comparison: the NIMBY benchmark and the fertility extension

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1 The model

This note takes the model section of Gross and Chivers (2025) as the baseline object and isolates the changes required to introduce fertility. The baseline NIMBY model is a life-cycle model with housing and debt choice. Households differ by age, liquid wealth, housing, and idiosyncratic income, and political support for additional housing supply is determined by the effect of a small, persistent change in house prices on expected lifetime utility.

The fertility extension preserves that structure. Real estate firms, the neighborhood voting problem, and the median-voter equilibrium condition remain in place. What changes is the household problem. The household state is augmented with family states, and the flow utility from housing is modified so that children at home reduce effective housing services through crowding.

1.1 Household's problem

Consider a household of age j with liquid assets b_t, housing a_t, and income state z_t. In the NIMBY benchmark, lifetime utility is

$$\sum_{j=0}^{J} \beta^j u(c_j, H_j) + \beta^{J+1} v(w_{J+1}), \tag{1}$$

where H_t denotes housing services from owner-occupied or rented housing and terminal utility is the same warm-glow bequest motive used in the baseline paper.

For the benchmark comparison considered here, the NIMBY flow utility is

$$u_t^N = \log c_t + s_h \log H_t - \text{penalty}_t, \tag{2}$$

with the terminal branch adding the bequest term

$$v(w_{t+1}) = \psi \log w_{t+1}. \tag{3}$$

The fertility extension adds two family states:

- $p_t \in \{0, 1, 2, 3+\}$, parity, or children ever born.
- $h_t \in \{0, 1, 2, 3\}$, children currently at home.

The state vector therefore changes from

$$\mathbf{s}_t^N = \{b_t, a_{t-1}, z_t\} \tag{4}$$

to

$$\mathbf{s}_t^F = \{b_t, a_{t-1}, z_t, h_t, p_t\}. \tag{5}$$

The key change is that housing enters utility through effective housing services rather than raw housing services. Let

$$H_t^{eff} = \frac{H_t}{(1+\lambda_c h_t)^{\psi_c}}. \tag{6}$$

Then, absent a birth in the current period, the fertility-extension flow utility is

$$u_t^{F,0} = \log c_t + s_h \log H_t^{eff} + \nu_h h_t - \text{penalty}_t. \tag{7}$$

If a birth occurs in the current period, the newborn counts immediately in both the crowding term and the child-utility term:

$$u_t^{F,1} = \log c_t + s_h \log\left(\frac{H_t}{(1+\lambda_c(h_t+1))^{\psi_c}}\right) + \nu_h(h_t+1) + \phi(p_t) - \kappa_0 - \kappa_q q_t - \text{penalty}_t. \tag{8}$$

Here $\phi(p_t)$ is parity-specific birth utility, κ_0 is the direct birth cost, and $\kappa_q q_t$ is the housing-price-sensitive birth cost.

Births therefore arise from a comparison between the birth and no-birth branches of the household problem.

The family-state transitions are

$$p_{t+1} = \min\{p_t + \mathbf{1}\{\text{birth}\}, 3+\}, \tag{9}$$

$$h_{t+1} = h_t + \mathbf{1}\{\text{birth}\} - \ell_t, \tag{10}$$

where $\ell_t \in \{0,1\}$ is a reduced-form leave-home shock. This separation between parity and children-at-home is the key modeling distinction in the corrected benchmark. A household's completed fertility is permanent, while the crowding state that governs housing demand is temporary.

The baseline NIMBY model is nested exactly when the fertility block is shut off. The precise parameter restrictions used for that nesting result are reported in the appendix.

1.2 Real estate firms

The real-estate block is unchanged. Rental housing is supplied competitively and the rental price is pinned down by the user cost of housing and expected capital gains. The zero-profit condition therefore remains

$$p_t^{rent} = \phi^{rent} + r_t^a p_t^h - \Delta p_{t+1}^h. \tag{11}$$

1.3 Housing supply

The local political mechanism is also unchanged. Households vote over a small, persistent change in local housing supply, taking the induced change in house prices as given. A household supports lower housing supply if a small increase in the house price raises expected lifetime utility:

$$V_j(\mathbf{s}; p^h + \epsilon^p) - V_j(\mathbf{s}; p^h) + \sigma \geq 0. \tag{12}$$

What changes is the object inside the vote calculation. In the fertility model the value function depends on both the housing portfolio and the household's family state, so the same price perturbation is filtered through an enlarged state space and a different utility function.

1.4 The median voter theorem

Applying the same median-voter logic as in the NIMBY benchmark, the equilibrium house price satisfies

$$\int g(\mathbf{s}; p_t^h)\Theta(\mathbf{s}; p^h)\, d\mathbf{s} = 0.5, \tag{13}$$

where

$$\Theta(\mathbf{s}; p^h) = \mathbf{1}_{\{V_j(\mathbf{s}; p^h+\epsilon^p) - V_j(\mathbf{s}; p^h) + \sigma \geq 0\}}. \tag{14}$$

The median-voter condition is therefore unchanged in form. The fertility extension matters because it shifts both the support function $\Theta(\cdot)$ and the stationary state distribution $g(\cdot)$.

1.5 Equilibrium

The political-economic equilibrium consists of household value functions, tenure and portfolio policies, births, the stationary distribution of households over the augmented state vector, and the house price that makes the median voter indifferent to a small change in supply. Relative to the baseline NIMBY benchmark, the fertility extension adds family-state policies and family-state distributions, but leaves the equilibrium concept itself unchanged.

2 Benchmark comparison

2.1 Exact nesting result

The first benchmark fact is exact nesting. When the fertility block is shut off, the fertility model reproduces the upstream NIMBY steady state exactly. The implementation details behind that verification are reported in the appendix.

Check	Gap
Distance gap	0
Vote gap	0
Debt gap	0

Table 1: Reproduction check: the fertility solver nests the NIMBY benchmark exactly in the shutoff case.

2.2 What changes in the benchmark

Using the common $rbPos = 0.03$ convention and tracing political support over the same house-price grid, the equilibrium crossing shifts materially once fertility is active.

The central quantitative result is that the fertility extension shifts the support schedule toward lower house prices. The equilibrium crossing moves from 2.326287 in the NIMBY benchmark to 1.751853 in the fertility benchmark. The interpretation is straightforward. In the baseline NIMBY model households care about house prices through portfolio and tenure considerations. In the fertility extension they also care about the effect of house prices on crowding and the cost of family formation. That additional margin makes high house prices less politically acceptable.

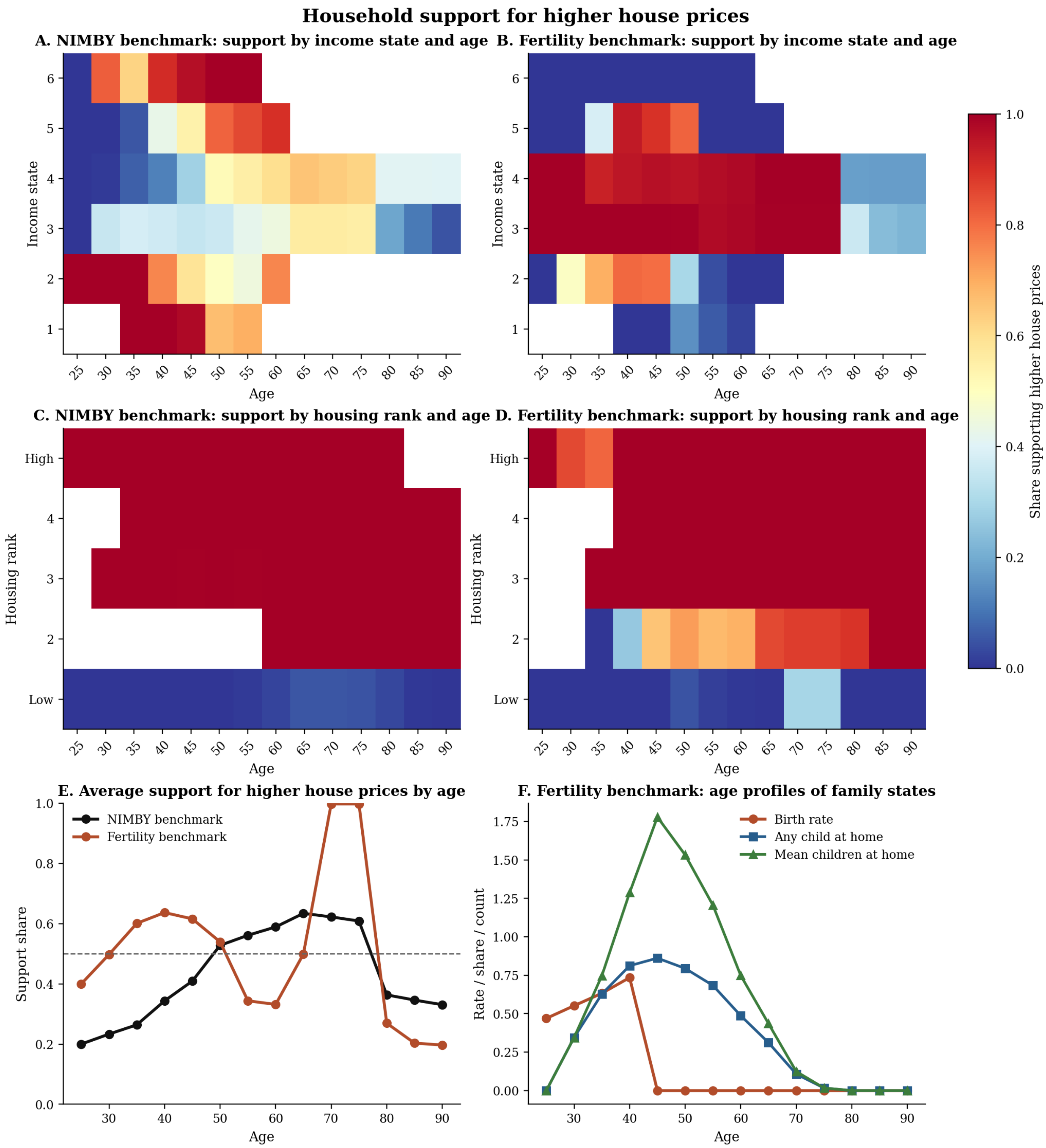

Figure 1: Household support for higher house prices. Panels A–D redraw the NIMBY paper's voting figures using the baseline NIMBY steady state and the fertility steady state. Panel E shows the age profile of support in the two benchmarks. Panel F adds the family-state profiles that are unique to the fertility extension.

Object	NIMBY benchmark	Fertility benchmark
Crossing house price	2.326287	1.751853
Vote at refined price	0.002227	0.008446
Vote per unit mass at refined price	0.000349	0.001370
Stationary household mass	6.387684	6.163346
Debt stock at refined price	-4.177670	44.406591
Rent share at refined price	3.963530	2.866941
Average birth rate	–	0.597710

Table 2: Steady-state benchmark comparison at the refined crossing implied by each model.

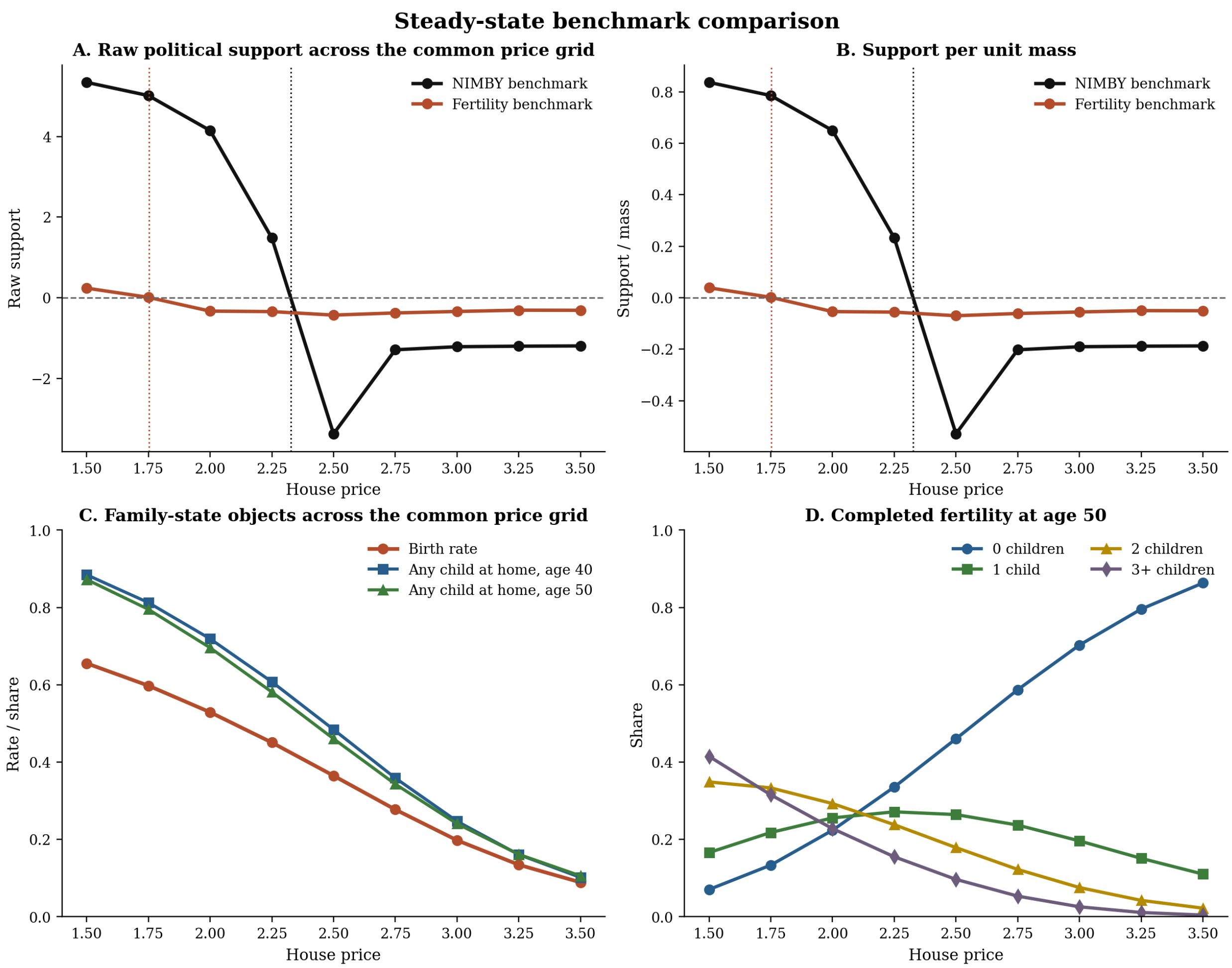

Figure 2: Steady-state benchmark comparison on the common house-price grid. The top row traces the raw vote schedule and the vote-per-mass schedule in the two models. The bottom row shows the family-state objects that move with house prices in the fertility benchmark.

3 Temporary birthrate increase

To connect the steady-state comparison to the original NIMBY paper's dynamic experiment, I impose the same temporary baby-boom shock used in the upstream transition exercise: a 10 percent increase in births for 10 periods. Figure 3 reports the resulting transition comparison. Panel A shows the common imposed shock. Panels B and C compare the demographic and house-price responses in the two models. Panels D and E add tenure-access responses, while Panel F isolates the additional family-state responses that only exist in the fertility extension.

The central transition result is that the fertility extension produces a larger medium-run house price response than the original NIMBY model under the same baby-boom shock. After normalizing each model's house-price path by its own initial level, the average post-window price deviation is about 2.0 percent in the NIMBY transition and about 5.1 percent in the fertility transition. At the same time, the fertility model displays an additional family adjustment that the NIMBY model does not contain: fertility rises during the boom by about 0.022 relative to baseline, then falls slightly below baseline in later periods, while the stock of children at home remains temporarily elevated.

The natural interpretation is that the baby boom does more than shift the age distribution. In the fertility model it also changes current family crowding and the demand for housing space. That extra family-demand channel amplifies the later tightening of the housing market relative to the original NIMBY transition.

Because the current project-03 transition block still abstracts from an explicit tenure choice, the homeownership panels in Figure 3 should be read as proxy objects. Each age bin is mapped into a common age-profile of homeownership, and that schedule is then shifted by the model-implied house-price path. Even with that conservative mapping, the fertility extension produces a much larger deterioration in young homeownership access than the original NIMBY transition. The trough in the young-homeownership response is about -0.0107 in the fertility transition, compared with about -0.0019 in the NIMBY benchmark. The same ranking does not hold for older households. The old-homeownership proxy in the fertility transition stays close to zero early on and reaches a much smaller late decline than the NIMBY line.

The original NIMBY paper also emphasizes cohort incidence. Figure 4 provides the closest current counterpart on the fertility side. The NIMBY lines are the upstream cohort homeownership indices for the boom generation, their parents, and their children. The fertility lines use the same age-and-price tenure mapping described above, evaluated along those cohort age paths. They are therefore access proxies, not structural cohort policy functions, but they are still informative about incidence. The parent generation is barely affected. The boom generation loses some homeownership access in midlife, and the child generation is hit most strongly, with the proxy index falling to roughly 0.974. That ordering matches the paper's intended mechanism: the stronger house-price response in the fertility model is borne most heavily by later cohorts trying to enter owner-occupation after the boom.

4 Historical fertility validation

A direct validation question is whether the fertility extension helps the baby-boom experiment match the subsequent decline in observed fertility. To check that, I use the aggregate historical weighted-birthrate series from the legacy NIMBY data and compare it to the NIMBY and fertility-extension transition paths. Figure 5 reports both the raw postwar series and the aligned post-boom comparison.

The important caveat is that the historical baby boom is more persistent than the toy 10-period

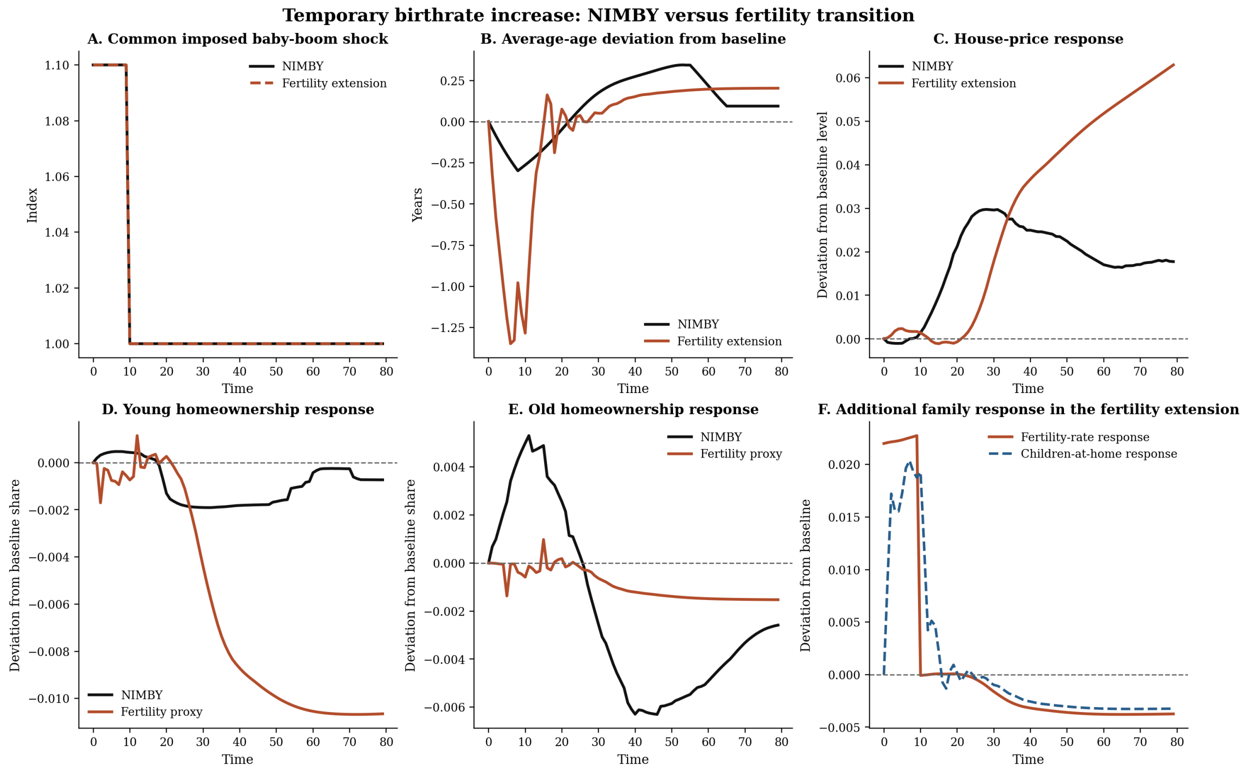

Figure 3: Temporary birthrate increase: NIMBY versus fertility transition. Panel A shows the common 10 percent baby-boom shock for 10 periods. Panel B compares the average-age response. Panel C compares the house-price response after normalizing each model by its initial house-price level. Panels D and E compare young and old homeownership responses; the fertility lines are composition-and-affordability tenure proxies because the current transition block does not solve an explicit tenure choice. Panel F reports the additional family-state responses that only appear in the fertility extension.

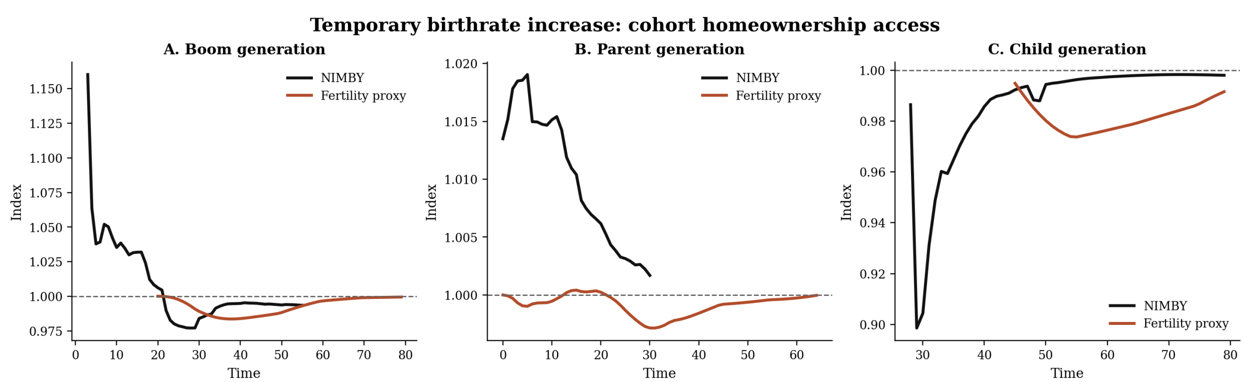

Figure 4: Temporary birthrate increase: cohort homeownership access. The black lines are the upstream NIMBY cohort homeownership indices. The red lines are fertility-side cohort homeownership-access proxies built from a common age profile and the model-implied house-price path. The comparison is informative about lifecycle incidence, but it is not yet a structural tenure comparison because the project-03 transition block does not solve household tenure choice.

shock used in the model. For that reason, the clean comparison is the shape of the decline after the boom window rather than exact calendar matching of the peak. Once the historical series is aligned at 1956 and the model is aligned at period 10, the fertility extension fits the decline modestly better than the NIMBY line at each horizon considered. Over the first 10, 20, 30, and 40 years of the aligned decline, the RMSE against the historical series is about 0.104, 0.177, 0.218, and 0.233 in the fertility extension, compared with 0.119, 0.183, 0.226, and 0.243 in the NIMBY path.

The interpretation is limited but useful. The original NIMBY transition has no mechanism that can push fertility below baseline after the imposed baby-boom window ends, so it returns immediately to its baseline level. The fertility extension adds that endogenous post-boom adjustment. The historical series displays a much larger and longer decline than either model, but the fertility extension moves the transition in the correct direction.

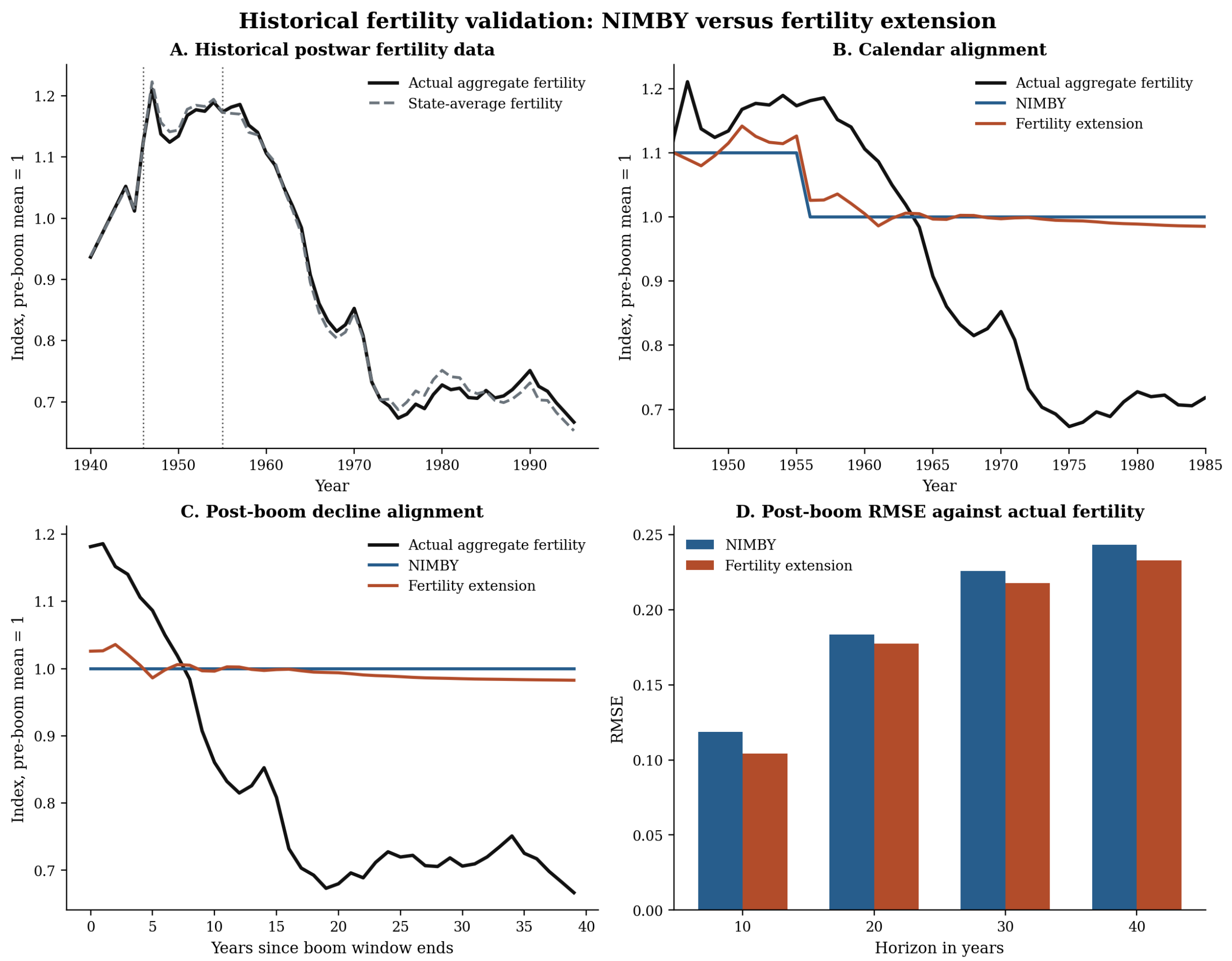

Figure 5: Historical fertility validation. Panel A plots the aggregate postwar fertility series from the legacy NIMBY data together with a state-average robustness series. Panel B calendar-aligns the historical series with the model paths, taking model period 0 as 1946. Panel C aligns the decline after the boom window, taking 1956 in the data and period 10 in the model as the common starting point. Panel D reports RMSEs for the aligned post-boom decline.

5 Robustness

The next question is whether the dynamic comparison is driven by a knife-edge parameter choice inside the bridge model. Figure 6 reports a bounded robustness sweep around three margins: the size of the baby-boom shock, the reduced-form price sensitivity of fertility, and the strength of family demand in the housing block. The NIMBY side of the sweep uses the project-03 proxy rather than the upstream IRF because only the benchmark 10 percent shock has been exported from the original NIMBY transition stack.

The main result is that the central price ranking is robust. Across all three shock sizes, the fertility extension produces a larger post-boom price response than the NIMBY proxy. At the benchmark 10 percent shock, the post-boom response is about 0.051 in the fertility extension against about 0.031 in the NIMBY proxy; at a 30 percent shock the same comparison becomes roughly 0.125 against 0.058. Varying the reduced-form price sensitivity of fertility between 0.10 and 0.30 hardly changes that ranking. Varying the family-demand strength shifts the level of the fertility price response more noticeably, but again the fertility line remains above the NIMBY proxy throughout.

The access results are more nuanced than the price results. Stronger shocks make young homeownership access worse in both models, and by the 30 percent shock the fertility extension has the more negative young-homeownership trough. But when the change comes from stronger price sensitivity of fertility or stronger family demand rather than a larger demographic shock, the fertility extension can partly self-correct through lower later births, so the young-access trough need not move monotonically with the price response. The robust conclusion is therefore the price amplification result, not a universal dominance ordering for every access proxy under every parameter perturbation.

6 Future demographic projections

The NIMBY paper also asks what future aging implies for house prices. The accessible upstream forecast files contain the age-weight scenarios used for that exercise, so Figure 7 feeds those same projected age weights into the current bridge model. The black lines shut off the fertility-demand channel and therefore play the role of the NIMBY side of the comparison. The red lines keep births and children at home in the demand block. This is still a bridge object rather than the full project-02 forecast solver, but it is the correct next comparison with the files currently available.

The projected-age results go in the opposite direction from the baby-boom impulse response. Under all three immigration scenarios, projected aging raises the NIMBY proxy price path but leaves the fertility path much flatter. In the medium-immigration scenario, the NIMBY proxy reaches a house price index of about 1.093 by 2050 and about 1.067 by 2100, while the fertility bridge is almost flat in 2050 at about 1.000 and slightly below its 2020 level by 2100 at about 0.976. The interpretation is that the family channel pushes in different directions in the short run and the long run. In the baby-boom experiment, a temporary wave of children at home raises contemporaneous space demand. In the long projection, aging lowers the mass of family-forming households, so the fertility channel dampens the aging-driven rise in NIMBY pressure rather than amplifying it.

Panel D of Figure 7 shows that the family-side projection objects remain quantitatively modest in the medium scenario. Relative to 2020, the fertility-rate index rises by roughly 2 percent in the 2020s, then settles close to one, while the young-homeownership proxy ends slightly above its 2020 level. So the main forecast message is not a future fertility collapse. It is that once the projected demographic path is dominated by population aging rather than a transitory child wave, the fertility extension tempers the long-run house-price increase implied by the baseline NIMBY mechanism.

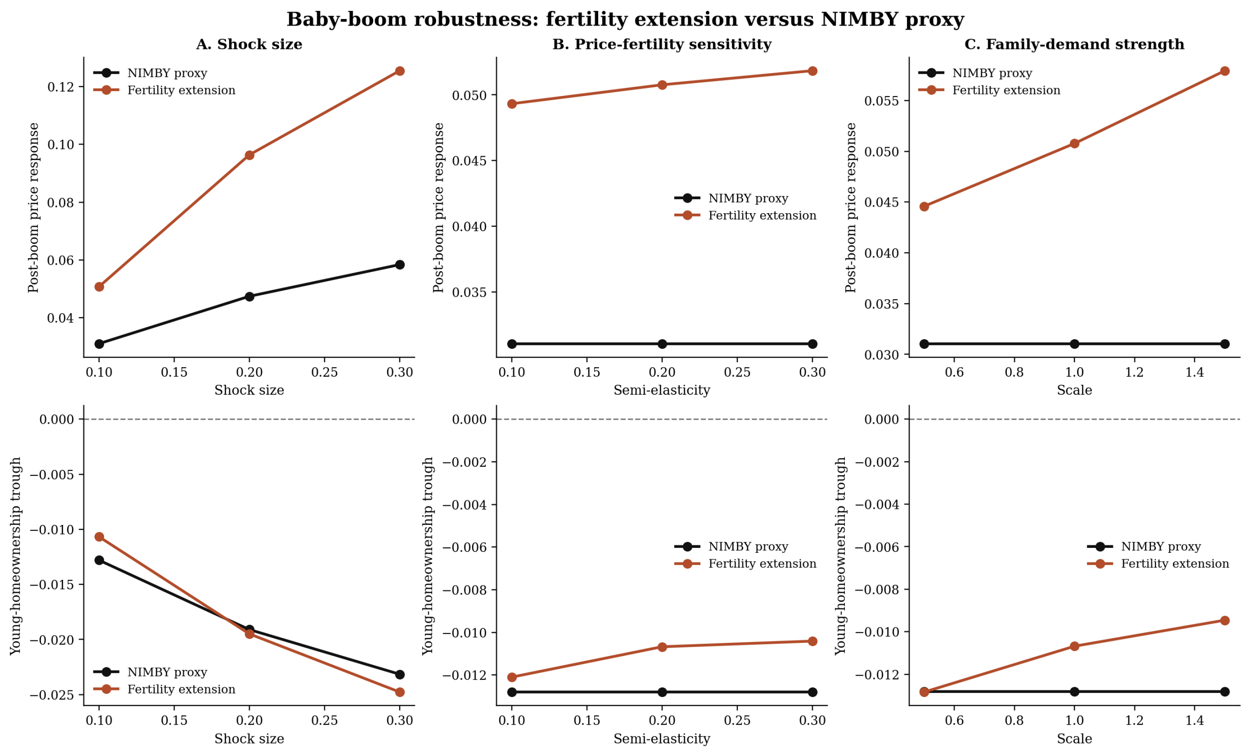

Figure 6: Baby-boom robustness. The top row reports the post-boom house-price response under alternative baby-boom sizes, reduced-form price sensitivities of fertility, and family-demand strengths. The bottom row reports the corresponding young-homeownership troughs. The black lines are the project-03 NIMBY proxy; the red lines are the fertility extension.

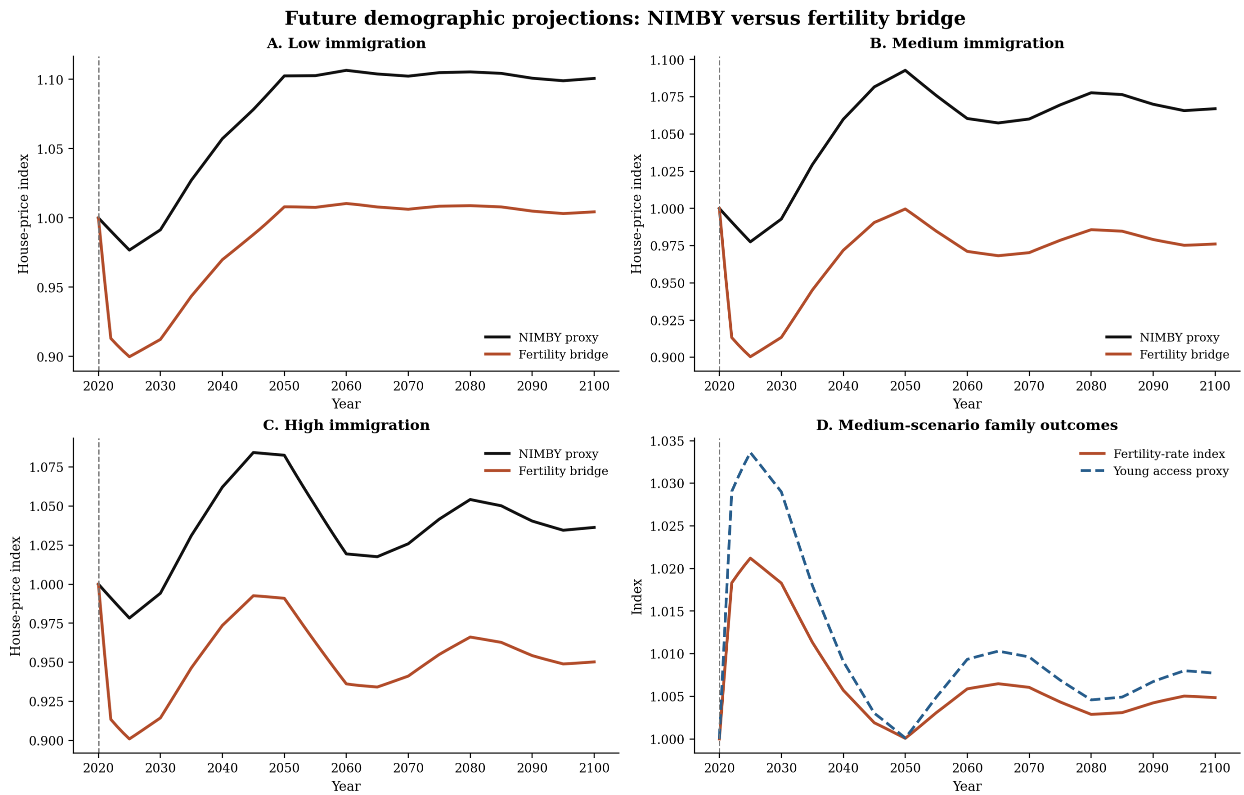

Figure 7: Future demographic projections. Panels A–C use the low-, medium-, and high-immigration age-weight scenarios from the upstream forecast files and compare the NIMBY proxy against the fertility bridge. Panel D reports the medium-scenario fertility-rate index and young-homeownership proxy. This is a matched projection bridge rather than the full project-02 forecast solver, but it uses the same upstream age scenarios on both sides of the comparison.

7 Summary

The comparison can be summarized in one sentence: the fertility model is not a different voting model, a different real-estate model, or a different equilibrium concept. It is the NIMBY model with an augmented household problem in which children at home reduce effective housing services and births depend on the value of that crowded housing state. Once that family margin is introduced, the political support schedule shifts toward lower house prices in the steady-state benchmark and amplifies the medium-run house-price response to a temporary baby boom. That price-amplification result is robust in the current bridge model. The historical validation exercise shows that the same fertility margin also moves the post-boom fertility path closer to the long decline in the aggregate US data, although the fit improvement is modest. The future-projection bridge adds one final contrast: when the demographic change is long-run aging rather than a temporary child wave, the fertility channel dampens the rise in projected house prices relative to the NIMBY proxy. The current transition and forecast blocks still omit the savings and net-worth objects behind the richer NIMBY cohort-incidence panels, but the basic comparison now covers steady states, baby-boom transitions, historical validation, bounded robustness, and forward demographic projections.

A Implementation and verification details

This note keeps implementation details out of the main text, but three technical points matter for the benchmark comparison.

First, in the numerical benchmark the birth and no-birth branches are mapped into a smooth birth probability using

$$\Pr(\text{birth} \mid \mathbf{s}_t) = \Lambda\left(\frac{V_t^{F,1}(\mathbf{s}_t; q_t) - V_t^{F,0}(\mathbf{s}_t; q_t)}{\varsigma}\right). \tag{15}$$

Second, exact nesting is obtained by shutting off the fertility block so that the fertility extension collapses to the NIMBY benchmark. In the benchmark implementation this means setting

$$\nu_h = \kappa_0 = \kappa_q = \lambda_c = \psi_c = 0 \tag{16}$$

and removing the active birth branch.

Third, the reported benchmark comparison uses the current steady-state benchmark objects:

- the fertility benchmark on the household grid $(I, J) = (60, 14)$;
- the upstream NIMBY benchmark on its native grid $(I, J) = (50, 10)$;
- a common price perturbation convention with $rbPos = 0.03$.

Fourth, the transition comparison uses the upstream NIMBY baby-boom reference from

`C:/Users/Dave_/Dropbox/Zac and David/Code/SteadyState/Mod_IRF/irfs_smoothed.mat`

because the smoothed IRF object is not stored in the repo copy. The fertility transition is generated from the current project-03 transition block under the same temporary 10 percent shock for 10 periods. House-price responses are reported as deviations from each model's own initial house-price level so the two transition lines are on a comparable scale.

Fifth, the historical validation uses the legacy aggregate fertility files

`C:/Users/Dave_/Dropbox/Zac and David/Data/merged_birthrates_migrationweights.dta`
`C:/Users/Dave_/Dropbox/Zac and David/Data/birthrates_updated.dta`

and normalizes both historical series by the mean pre-boom birth rate over 1940–1945. The main comparison is a post-boom shape comparison, not a formal estimation exercise.

Sixth, the robustness sweep is implemented through a Python bridge that reproduces the benchmark MATLAB policy runs to machine precision before varying the shock size, price sensitivity, and family-demand strength. The NIMBY side of that sweep is the project-03 proxy rather than a fresh set of upstream IRFs.

Seventh, the future-demographic projection comparison uses the upstream forecast age weights from `C:/Users/Dave_/Dropbox/ZacandDavid/Code/SteadyState/Mod_DynamicForecast/loop output_forecast.mat` but the reported comparison is a matched bridge exercise rather than the full project-02 forecast solver. The black line in Figure 7 is the NIMBY proxy with the fertility-demand channel shut off; the red line adds births and children-at-home demand under the same projected age path.

Finally, the transition comparison should be read as dynamic mechanism evidence rather than as a fully unified household-DP transition benchmark. The steady-state comparison is the exact household-to-household comparison; the transition section shows how the fertility mechanism alters the original baby-boom experiment under the current project-03 dynamic block. The richer homeownership objects from the original NIMBY IRFs are now matched by reduced-form composition-and-affordability proxies on the fertility side, but the transition block still does not solve a savings problem and therefore does not yet generate directly comparable fertility-side net-worth panels.